

ICD 1.9a/3.1

“Near IR Imager (NIRI) to Observatory Control System”

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Hubert Yamada

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Chapter 1 Introduction

1-1.0 Description

1-1.1 Purpose

This Interface Control Document (ICD) serves the following purposes:

- To describe the commands and parameters that will pass between the OCS and the Near Infra-Red Imager (NIRI) sequencer and subsystems.

This document only describes the ways in which NIRI differs from the prototypical Gemini instrument, described in ICD 1.9/3.1 [16]. It is assumed that the reader is familiar with that document. *Much of this document is copied verbatim from ICD 1.9/3.1, without further acknowledgement.*

- To provide the necessary declarations of commands, parameters and Status and Information records which is required in an “Instrument to OCS ICD.” This document provides the required tables, using the required “Parameter Definition Format”, described in [11].

In addition to the documents mentioned in the previous paragraphs, the specific interfaces described in this document are governed by the following parent Interface Control Documents, which describe the general properties of particular kinds of interface:

- ICD 1a [24], which describes how the OCS deals with commands and attributes and the behavior the OCS expects from the systems it commands.
- ICD 1b [25], which gives a general description of the way the Observatory Control System communicates with the other principal systems. It is also related to ICD 7a, [27], which describes the way an Instrument Control System communicates with its subsystems.
- ICD 2 [26], which describes the way in which an Instrument Control System makes information available to the outside world (and the OCS) by means of “Status Information Records”.

The intended audience for this document is:

- the NIRI reviewers;
- the developers of the Gemini Observatory Control System (OCS).

1-1.2 Scope

This document describes only aspects of the NIRI software system which differ from, or are more specific than, the Gemini standard instrument, described in [16]. It is assumed that the reader is familiar with that documents.

This document does *not* describe the following topics:

- the science detector, except for sequence commands which will be forwarded to the science detector controller (see [18]);
- the NIRI hardware and electronics (see [11] and [12], summarized in [3]);

- the NIRI On Instrument Wavefront Sensor (see [9]).

The software design is not described here. For that, see [2] (currently under preparation), and [3] (currently under preparation).

1–2.0 Related Documents and Drawings

1–2.1 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1–2.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1–2.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

Chapter 2 Details of the NIRI to OCS Interface

2-1.0 Introduction

2-2.0 Physical System Interfaces

2-2.1 Mechanical Interface

Not Applicable for a software ICD.

2-2.2 Optical Interface

Not Applicable for a software ICD.

2-2.3 Electronic Interface

The system hardware architecture for the NIRI control software is the same as described in [16], with the following exception:

- The Instrument Sequencer, Components Controller, and OIWFS Components Controller will share the same IOC.

There will also be the following electronic VME modules:

- The XVME-240 Digital I/O Board, (shared between the NIRI CC, the NIRI OIWFS, and the interlock bus):
 - To support the interlock bus (Gemini standard, see [29]);
 - To monitor the NIRI and NIRI OIWFS stepper-motor control modules;
 - To reset the NIRI and NIRI OIWFS stepper-motor control modules;
 - To control the heating system.
- A XVME 490 four-port serial I/O board:
 - To control the NIRI/NIRI OIWFS temperature controllers (three ports);
 - To control the NIRI/NIRI OIWFS temperature sensor (one port);
- Four custom-built VME sensor-support boards:
 - To read out the NIRI Hall-effect sensors;
 - To read out the NIRI OIWFS Hall-effect sensors;
- Two XVME-566 Analog Input Modules
 - To read out the VME sensor-support boards for the NIRI Hall-effect sensors;
 - To read out the VME sensor-support boards for the NIRI OIWFS Hall-effect sensors;
- Three Oregon Microsystems VME8-8 Stepper-motor controller boards
 - To control the NIRI stepper-motors
 - To control the NIRI OIWFS stepper-motors;

2–2.4 Mass/Balance*Not Applicable for a software ICD.***2–2.5 Thermal Interface***Not Applicable for a software ICD.*

2–3.0 Software/Control Function Interface

2–3.1 Overview**2–3.1.1 Communications infrastructure**

The communications infrastructure is the same as described in [16].

2–3.1.2 Software architecture

This section is not yet written.

2–3.2 Behavior

The behavior of the NIRI/OCS interface is the same as described in [16].

2–3.3 Implementation

This section contains tables summarizing the EPICS records used for the interface. The next sections (Section 2–3.4 on page 18 and Section 2–3.5 on page 24) contain the details of what those records do.

It is assumed in all cases that the record names given in this document will be prefixed by the string “niri:” when the database is implemented. For the current (prototype) release, records in the SAD database are prefixed by “niri::”; this will change to “niri:” before the final release.

2–3.3.1 Command (CAD/CAR) records

The columns of the command record tables are as follows:

Description. A brief description of the command.

Attribute. The name of the attribute which the Observatory Control System associates with a particular parameter of a particular command.

CAD record/field. The name of the EPICS CAD record (minus the “niri:” prefix¹) together with the field in that CAD record associated with each parameter of the command.

CAD ordering number. The recommended order in which this command should be executed. CAD records should be connected to their associated APPLY record in the order 1, 2, 3, 4, 5, etc... CAD records within the same ordering number set can be connected to the APPLY record in any order in that set.

1. Note that the Detector Controller commands have a “niri:dc:” prefix; the Components Controller commands have a “niri:cc:” prefix; and the environment commands have a “niri:env:” prefix.

CAR record. The CAR record associated with the command. Note that several CAD records may share the same CAR record.

2-3.3.1.1 Sequence command CAD/CAR records

The NIRI sequencer recognizes all the OCS sequence commands and uses the CAD/CAR records specified in ICD 1b [25]. See also ICD 1a [24], for a description of what these commands do.

TABLE 1. NIRI sequence command CAD/CAR records

Description	Attribute	CAD ("niri:" +)	CAR record ("niri:" +)
Test.	(none) ^a	test	testC ^b
Initialize.	(none) ^a	init	initC
Datum mechanisms.	(none) ^a	datum	datumC
Verify	(none) ^a	verify	verifyC
End verify	(none) ^a	endVerify	endVerifyC
Guide	(none) ^a	guide	guideC
End guide	(none) ^a	endGuide	endGuideC
Observe	(none) ^a	observe	observeC
Pause	(none) ^a	pause	
Continue	(none) ^a	continue	
Stop	(none) ^a	stop	
Abort	(none) ^a	abort	
End Observe	(none) ^a	endObserve	endObserveC
Park	(none) ^a	park	parkC
Diagnose	(none) ^a	diagnose	applyC ^c

- No arguments.
- Currently, it is required to have separate TestC, InitC, etc. CAR records. There some disagreement over this requirement, and all of these may CAR records may disappear, and will be replaced by the global applyC.
- This is not a standard record, and is not required to have its own diagnoseC CAR record.

2-3.3.1.2 General observing command CAD/CAR records

TABLE 2. NIRI general observing command CAD/CAR records

Description	Attribute	CAD record/field ("niri:" +)	CAD ordering number	CAR record ("niri:" +)
Select beam-splitter position.	Position name	cc:spltSel.A	6	cc:spltC
Park beam-splitter.	(none) ^a	cc:spltPark	14	
Select filter in wheel 1.	Position name	cc:filt1Sel.A	6	cc:filt1C
Park filter wheel 1.	(none) ^a	cc:filt1Park	14	
Select filter in wheel 2.	Position name	cc:filt2Sel.A	6	cc:filt2C
Park filter wheel 2.	(none) ^a	cc:filt2Park	14	
Select pupil mask or filter in pupil-mask/filter wheel.	Position name	cc:filt3Sel.A	6	cc:filt3C
Park pupil-mask/filter wheel.	(none) ^a	cc:filt3Park	14	
Select focal-plane mask.	Position name	cc:foplSel.A	6	cc:foplC
Park focal-plane mask wheel.	(none) ^a	cc:foplPark	14	
Select beam-steerer positions	Position name	cc:sterSel.A	6	cc:sterC
Park beam-steerers.	(none) ^a	cc:sterPark	14	
Select pupil-viewer position	Position name	cc:puvwSel.A	6	cc:puvwSelC
Park pupil-viewer.	(none) ^a	cc:puvwPark	14	
Set focus.	Focus position	cc:focMove.A	6	cc:focC
Park focus.	(none) ^a	cc:focPark	14	
Open/Close window cover	State [OPEN CLOSED]	cc:covSel.A	6	cc:covC
Park window cover. ^b	(none) ^a	cc:covPark	14	

a. No arguments.

b. Equivalent to closing the window cover.

2-3.3.1.3 Engineering command CAD/CAR records—System Commands

The following commands are used for engineering purposes, and are not expected to be used during normal observing. These are normally activated by the engineering user interface.

These commands affect the overall behavior of NIRI and the NIRI OIWFS. Note, in particular, that it is not possible to reboot NIRI and the NIRI OIWFS separately

TABLE 3. NIRI system engineering command CAD/CAR records

Description	Attribute	CAD record/field ("niri:" +)	CAD ordering number	CAR record
Reboot.	(none) ^a	reboot	N/A	N/A

TABLE 3. NIRI system engineering command CAD/CAR records

Description	Attribute	CAD record/field ("niri:" +)	CAD ordering number	CAR record
Select debug mode.	Debug mode	debug.A	3	debugC
Flush data from internal buffers.	(none) ^a	flush	13	flushC

a. No arguments.

2-3.3.1.4 CC Sequence command CAD/CAR records

The NIRI components controller control software recognizes all the OCS sequence commands and uses the CAD/CAR records specified in ICD 1b, [25]. See also ICD 1a, [24], for a description of what these commands do.

TABLE 4. NIRI CC sequence command CAD/CAR records

Description	Attribute	CAD record.FIELD ("niri:" +)	CAR record ("niri:" +)
Test.	(none)	cc:test	cc:testC
Initialize.	(none)	cc:init	cc:initC
Datum mechanisms.	(none)	cc:datum	cc:datumC
Verify	(none)	cc:verify	cc:verifyC
End verify (no action)	(none)	cc:endVerify	cc:endVerifyC
Guide	(none)	cc:guide	cc:guideC
End guide (no effect)	(none)	cc:endGuide	cc:endGuideC
Park	(none)	cc:park	cc:parkC

2-3.3.1.5 Engineering command NIRI CC CAD/CAR records

The following commands are used for engineering purposes, and are not expected to be used during normal observing. Note that, although these commands are connected to the APPLY record, they are normally activated directly by the engineering user interface.

The following commands affect the overall behavior of NIRI.

TABLE 5. NIRI CC engineering command CAD/CAR records

Description	Attribute	CAD record/field ("niri:" +)	CAD ordering number	CAR record ("niri:" +)
Datum filter wheel 1.	(none) ^a	cc:flt1Datm	3	cc:flt1C
Set filter wheel 1 position in engineering units ^b .	Filter wheel 1 position	cc:flt1EngMove.A	6	
Confirm that filter wheel 1 has not lost its datum position.	(none) ^a	cc:flt1Rdtm	6	
Confirm that filter wheel 1 and its Hall-effect sensors are operational.	(none) ^a	cc:flt1Diag	6	
Datum filter wheel 2.	(none) ^a	cc:flt2Datm	3	cc:flt2C
Set filter wheel 2 position in engineering units ^b .	Position	cc:flt2EngMove.A	6	
Confirm that filter wheel 1 has not lost its datum position.	(none) ^a	cc:flt2Rdtm	6	
Confirm that filter wheel 2 and its Hall-effect sensors are operational	(none) ^a	cc:flt2Diag	6	
Datum pupil-mask/filter wheel (filter wheel 3).	(none) ^a	cc:flt3Datm	3	cc:flt3C
Set pupil-mask/filter wheel (filter wheel 3) position in engineering units ^b .	Position	cc:flt3EngMove.A	6	
Confirm that pupil-mask/filter wheel (filter wheel 3) has not lost its datum position.	(none) ^a	cc:flt3Rdtm	6	
Confirm that pupil-mask/filter wheel and its Hall-effect sensors are operational	(none) ^a	cc:flt3Diag	6	
Datum focal-plane mask wheel.	(none) ^a	cc:foplDatm	3	cc:foplC
Set focal-plane mask wheel position in engineering units ^b .	Focal-plane mask wheel position	cc:foplEngMove.A	6	
Confirm that focal-plane mask wheel has not lost its datum position.	(none) ^a	cc:foplRdtm	6	
Confirm that focal-plane mask wheel and its Hall-effect sensors are operational	(none) ^a	cc:foplDiag	6	

TABLE 5. NIRI CC engineering command CAD/CAR records

Description	Attribute	CAD record/field ("niri:" +)	CAD ordering number	CAR record ("niri:" +)
Datum beam steerers.	(none) ^a	cc:sterDatm	3	cc:sterC
Set beam steerer 1 position in engineering units ^b .	Motor pos. (long)	cc:ster1EngMove.A	6	
Confirm that beam steerer 1 has not lost its datum position.	(none) ^a	cc:ster1Rdtm	6	
Confirm that beam steerer 1 and its Hall-effect sensors are operational	(none) ^a	cc:ster1Diag	6	
Set beam-steerer 2 position in engineering units ^b .	Motor pos. (long)	cc:ster2EngMove.A	6	
Confirm that beam steerer 2 has not lost its datum position.	(none) ^a	cc:ster2Rdtm	6	
Confirm that beam steerer 2 and its Hall-effect sensors are operational	(none) ^a	cc:ster2Diag	6	
Datum pupil-viewer mechanism	(none) ^a	cc:puvwDatm	3	cc:puvwC
Set pupil-viewer position in engineering units ^c .	Position (long)	cc:puvwEngMove.A	6	
Confirm that pupil viewer and its Hall-effect sensors are operational	(none) ^a	cc:puvwDiag	6	
Datum focus mechanism	(none) ^a	cc:focDatm	3	cc:focC
Set focus translation mechanism in engineering units.	Position (long)	cc:focEngMove.A	6	
Confirm that focus stage has not lost its datum position.	(none) ^a	cc:focRdtm	6	
Confirm that focus stage and its Hall-effect sensors are operational	(none) ^a	cc:focDiag	6	
Datum window cover.	(none) ^a	cc:covDatm	3	cc:covC
Set window cover position in engineering units.	Position (long)	cc:covEngMove.A	6	
Confirm that window cover has not lost its datum position.	(none) ^a	cc:covRdtm	6	
Confirm that window cover and its Hall-effect sensors are operational	(none) ^a	cc:covDiag	6	

a. No arguments.

b. Arbitrary locations are not fully repeatable.

c. *This is strongly discouraged.* The only stable locations are at the predefined locations. The predefined locations are only stable when properly approached.

2-3.3.2 Status Information Records

The following tables show the status information provided by NIRI, together with the subset of that information which is written to the FITS header. The tables distinguish status information provided by the Instrument Sequencer, Detector Controller and Components Con-

troller, but all the information shown is available over the OCS to NIRI interface. The tables have the following columns:

SIR record. The part of the name of the EPICS SIR record which follows the “niri:” prefix².

Units. The units in which the quantity is stored

Comments. A description of the item. The first paragraph of this will go into the SIR record description and also the comment field in the FITS header.

Note that only the type, units and comments are actually contained in the SIR records. The FITS keyword is provided by the DHS.

TABLE 6. NIRI Instrument Sequencer (IS) status Information

SIR record (“niri:” +)	FITS keyword	FITS included?	Type	Units	Comments
name	INSTRUME	Start	string		Instrument name (“NIRI”).
state	INSTATE		string		Instrument state [BOOTING INITIALISING RUNNING CONFIGURING]
health	INHEALTH		string		Instrument health [GOOD WARNING BAD].
issPort	INPORT		integer		Number of ISS port on which the instrument is installed.
historyLog		TBD	char array		Instrument log message.
heartBeat	N/A	Never	integer		Continuously changing variable used to detect whether the system is still alive.
simMode	N/A	Never	string		Simulation mode [NONE FULL FAST ^a]

- a. The FAST simulation mode removes most of the simulated delays which are built into the FULL simulation mode, but does not meet the Gemini specification for the FAST simulation mode. The VSM mode is not available.

2. Note that the actual system has been designed so the Status Alarm Database can easily be given a different prefix from the rest of the NIRI database (for example “niri:”). It is assumed in this document that the Status Alarm Database prefix will be “niri:”.

integer (0=YES,1=NO)The next table contains the same information as the previous on, but for the Components Controller, not the sequencer.

TABLE 7. NIRI Components Controller (CC) status Information

SIR record ("niri:" +)	FITS keyword	FITS included?	Type	Units	Comments
cc:name	CCNAME		string		Components Controller name ("NIRICC").
cc:state	CCSTATE		string		Components Controller state [BOOTING INITIALISING RUNNING CONFIGURING].
cc:health	CCHEALTH		string		Components Controller health [GOOD WARNING BAD].
cc:historyLog		TBD	char array		Components Controller log message.
cc:heartBeat	N/A	Never	integer		Continuously changing variable used to detect whether the system is still alive.
cc:simMode	N/A	Never	string		Components Controller Simulation mode [NONE FULL FAST ^a]
cc:wfsBeam	WFSBEAM		integer (0=YES 1=NO)		Flag to indicate that the NIRI is allowing light to fall on its on-instrument wavefront sensor.
cc:flt1Name	CCFLT1	Start	string		Filter 1 name.
cc:flt1Eng	CCFLT1EN		integer	engineering units	Filter wheel 1 engineering position.
cc:flt1Parked			integer (0=YES 1=NO)		Is filter wheel 1 parked?
cc:flt1Datumed	CCFLT1DT		integer (0=YES 1=NO)		Is filter wheel datumed?
cc:flt1Health	CCFLT1HT		string		Health of filter wheel 1.
cc:flt1PosErr			string (GOOD WARNING)		Filter wheel 1 properly positioned?
cc:flt1Module			string (GOOD ERROR)		Health of filter wheel 1 driver module.
cc:flt1Hallstat			string (GOOD WARNING)		Filter wheel 2 Hall effect sensors enabled?
cc:flt1State	CCFLT1ST		string		State of filter wheel 1.
cc:flt2Name	CCFLT2 •	Start	string		Filter 2 name.
cc:flt2Eng	CCFLT2EN		integer	engineering units	Filter wheel 2 engineering position.

TABLE 7. NIRI Components Controller (CC) status Information

SIR record ("niri:" +)	FITS keyword	FITS included?	Type	Units	Comments
cc:filt2Parked			integer (0=YES 1=NO)		Is filter wheel 2 parked?
cc:filt2Datumed	CCFLT2DT		integer (0=YES 1=NO)		Is filter wheel datumed?
cc:filt2Health	CCFLT2HT		string		Health of filter wheel 2.
cc:filt2PosErr			string (GOOD WARNING)		Filter wheel 2 properly positioned?
cc:filt2Module			string (GOOD ERROR)		Health of filter wheel 1 driver module.
cc:filt2Hallstat			string (GOOD WARNING)		Filter wheel 2 Hall effect sensors enabled?
cc:filt2State	CCFLT2ST		string		State of filter wheel 2.
cc:filt3Name	FILTER3	Start	string		Pupil-mask/filter name.
cc:filt3Eng	CCFLT3EN		integer	engineering units	Pupil-mask/filter wheel engineering position.
cc:filt3Parked			integer (0=YES 1=NO)		Is pupil-mask/filter wheel parked?
cc:filt3Datumed	CCFLT3DT		integer (0=YES 1=NO)		Is pupil-mask/filter wheel datumed?
cc:filt3Health	CCFLT3HT		string		Health of pupil-mask/filter wheel.
cc:filt3State	CCFLT3ST		string		State of pupil-mask/filter wheel.
cc:filt3PosErr			string (GOOD WARNING)		Pupil-mask/filter wheel properly positioned?
cc:filt3Module			string (GOOD ERROR)		Health of pupil-mask/filter wheel driver module.
cc:filt3Hallstat			string (GOOD WARNING)		Pupil-mask/ filter wheel all sensors enabled?
cc:sterName	CCSTR	Start	string		Beam steerer position name ^b .
cc:ster1Eng	CCSTR1FE		integer	engineering units	Set beam steerer 1 engineering position.
cc:ster1Parked			integer (0=YES 1=NO)		Is beam steerer 1 parked?

TABLE 7. NIRI Components Controller (CC) status Information

SIR record ("niri:" +)	FITS keyword	FITS included?	Type	Units	Comments
cc:ster1Datumed	CCSTR1DT		integer (0=YES 1=NO)		Is beam steerer 1 datumed?
cc:ster1Health	CCSTR1HT		string		Health of beam steerer 1.
cc:ster1PosErr			string (GOOD WARNING)		Beam steerer 1 properly posi- tioned?
cc:ster1Module			string (GOOD ERROR)		Health of beam steerer 1 driver module.
cc:ster1Hallstat			string (GOOD WARNING)		Beam steerer 1 Hall effect sen- sors enabled?
cc:ster1State	CCSTR2ST		string		State of beam steerer 1.
cc:ster2Eng	CCSTR2EN		integer	engineering units	Beam steerer 2 engineering position.
cc:ster2Parked			integer (0=YES 1=NO)		Is beam steerer 2 parked?
cc:ster2Datumed	CCSTR2DT		integer (0=YES 1=NO)		Is beam steerer 2 datumed?
cc:ster2Health	CCSTR2HT		string		Health of beam steerer 2.
cc:ster2PosErr			string (GOOD WARNING)		Beam steerer 2 properly posi- tioned?
cc:ster2Module			string (GOOD ERROR)		Health of beam steerer 2 driver module.
cc:ster2Hallstat			string (GOOD WARNING)		Beam steerer 2 all sensors enabled?
cc:ster2State	CCSTR2ST		string		State of beam steerer 2.
cc:foplName	CCFPM	Start	string		Focal-plane mask wheel name.
cc:foplEng	CCFPMEN		integer	engineering units	Focal-plane mask wheel engi- neering position.
cc:foplParked			integer (0=YES 1=NO)		Is focal-plane mask wheel parked?
cc:foplDatumed	CCFPMDT		integer (0=YES 1=NO)		Is focal-plane mask wheel datumed?
cc:foplHealth	CCFPMHT		string		Health of focal-plane mask wheel.

TABLE 7. NIRI Components Controller (CC) status Information

SIR record ("niri:" +)	FITS keyword	FITS included?	Type	Units	Comments
cc:foplPosErr			string (GOOD WARNING)		Focal-plane mask wheel prop- erly positioned?
cc:foplModule			string (GOOD ERROR)		Health of focal-plane mask wheel driver module.
cc:foplHallstat			string (GOOD WARNING)		Focal-plane mask wheel Hall effect sensors enabled?
cc:foplState	CCFPMST		string		State of focal-plane mask wheel.
cc:covEng	CCCOVEN		integer	engineering units	Window cover engineering posi- tion.
cc:covParked			integer (0=YES 1=NO)		Is window cover parked?
cc:covHealth	CCCOVHT		string		Health of window cover.
cc:coverPosErr			string (GOOD WARNING)		Cover properly positioned?
cc:coverModule			string (GOOD ERROR)		Health of cover driver module.
cc:coverHallstat			string (GOOD WARNING)		Cover Hall effect sensors enabled?
cc:covState	CCCOVST		string		State of window cover.
cc:pupvwName	CCPV	Start	string		Pupil viewer position name.
cc:pupvwEng	CCPVEN		integer	engineering units	Pupil viewer engineering posi- tion.
cc:pupvwParked			integer (0=YES 1=NO)		Is pupil viewer parked?
cc:pupvwHealth	CCPVHT		string		Health of pupil viewer.
cc:pupvwPosErr			string (GOOD WARNING)		Pupil viewer properly posi- tioned?
cc:pupvwModule			string (GOOD ERROR)		Pupil viewer driver module.
cc:pupvwHallstat			string (GOOD WARNING)		Pupil viewer Hall effect sensors enabled?
cc:pupvwState	CCPVST		string		State of pupil viewer.
cc:focName	CCFOC	Start	string		Focus position name ^c .

TABLE 7. NIRI Components Controller (CC) status Information

SIR record ("niri:" +)	FITS keyword	FITS included?	Type	Units	Comments
cc:focParked			integer (0=YES 1=NO)		Is focus stage parked?
cc:focDatumed	CCFOCDT		integer (0=YES 1=NO)		Is focus stage datumed?
cc:focHealth	CCFOCHT		string		Health of focus stage.
cc:focPosErr			string (GOOD WARNING)		Focus stage properly posi- tioned?
cc:focModule			string (GOOD ERROR)		Health of focus stage driver module.
cc:focHallstat			string (GOOD WARNING)		Focus stage Hall effect sensors enabled?
cc:focState	CCFOCST		string		State of focus stage.
cc:focEng	CCFOCEN		integer	engineering units	Focus stage engineering posi- tion.
cc:focPos	CCFOCPOS		float	mm	Focus stage position (millime- ters)

- The FAST simulation mode removes most of the simulated delays which are built into the FULL simulation mode, but does not meet the Gemini specification for the FAST simulation mode. The VSM mode is not available.
- The positions of the individual beam steerers do not have individual names, because the beam steerers must always be moved as a pair.
- Since this is a continuous mechanism, the position may be "INVALID," meaning that it is not at a predefined position. This is not an error.

TABLE 8. NIRI Environment Information

SIR record ("niri:" +)	FITS keyword	FITS included?	Type	Units	Comments
env:health	CEHEALTH		string		Components Controller environ- ment health [GOOD WARN- ING BAD].
env:tcsp1Set	CETPSET1		float	K	Temperature controller 1 set point.
env:tct1Temp	CETPVAL1		float	K	Temperature controller 1 tem- perature
env:tcsp2Set	CETPSET2		float	K	Temperature controller 2 set point.
env:tct2Temp	CETPVAL2		float	K	Temperature controller 2 tem- perature

TABLE 8. NIRI Environment Information

SIR record ("niri:" +)	FITS keyword	FITS included?	Type	Units	Comments
env:dewarTempK1	CETPMON1		float	K	Dewar temperature, sensor 1.
env:dewarTempR1	CETPRAW1		float	V	Dewar temperature, sensor 1, raw voltage.
env:dewarTempK2	CETPMON2		float	K	Dewar temperature, sensor 2.
env:dewarTempR2	CETPRAW2		float	V	Dewar temperature, sensor 2, raw voltage.
...	...	...	...	...	...
env:dewarTempKn ^a	CETPMONn		float	K	Dewar temperature, sensor n.
env:dewarTempRn	CETPRAWn		float	V	Dewar temperature, sensor n, raw voltage.

a. n = total number of temperature sensors, currently TBD.

2–3.4 Detailed Command Description

A general table of the commands accepted by the NIRI control software, together with their arguments, is given in Table 2 on page 8. This section contains a detailed description of each of the commands that will be sent between the OCS and the NIRI control software. It is based on the required behavior of the commands specified in [21] and [32].

Any directive asserted on the top-level APPLY record will be forwarded to all the CAD records in the sequencer, CC, and DC (perhaps via lower-level APPLY records). Only the marked CAD records will respond. If the directive is accepted it will cause the NIRI control software to change its configuration. The “applyC” CAR record will go “BUSY” while the NIRI control software is changing its configuration and go to “IDLE” if completed successfully or to “ERR” if the configuration change fails.

2–3.4.1 Sequence Commands

2–3.4.1.1 TEST

On receipt of this command, the NIRI IS will insure that an observation is not taking place, and then will forward this command to the DC and CC. The “testC” CAR record will go “BUSY” while the tests are being carried out and go to “IDLE” if completed successfully or to “ERR” if the tests fail.

The CC accepts, but ignores this command.

The behavior of the DC is documented in [18].

Because the TEST command is not permitted to move any mechanisms, it is unable to detect any problems with the components controller. Use the DIAGNOSE and REDATUM commands instead.

2-3.4.1.2 INIT

On receipt of this command, the NIRI IS will insure that an observation is not taking place. It will then execute its initialization sequence, in which it will: Read its hardware setup files and lookup tables, and reset itself to the start-up state. This command will be forwarded to the CC and the DC. The “initC” CAR record will go “BUSY” while the initialization is being carried out and go to “IDLE” if completed successfully or to “ERR” if the initialization fails.

The CC will read its hardware setup files and lookup tables, and reset itself to the start-up state. No mechanisms will be moved.

The behavior of the DC is documented in [18].

2-3.4.1.3 DATUM

On receipt of this command, the NIRI IS will insure that an observation is not taking place and then forward the command to the CC and DC. The “datumC” CAR record will go “BUSY” while the datum search is being carried out and go to “IDLE” if completed successfully or to “ERR” if the datum searched fails.

The CC will locate the datum (i.e. index, home or reference point) for all its mechanisms.

The behavior of the DC is documented in [18].

2-3.4.1.4 VERIFY

On receipt of this command, the NIRI IS will insure that an observation is not taking place and then forward the command to the CC and DC. The “verifyC” CAR record will be changed to “BUSY”. The command will complete, and the “verifyC” CAR changed back to “IDLE” when the state of the NIRI control software can be changed safely.

The CC accepts, but ignores this command.

The behavior of the DC is documented in [18].

2-3.4.1.5 ENDVERIFY

On receipt of this command, the NIRI IS will insure that an observation is not taking place and then forward the command to the CC and DC. The command will change the “endVerifyC” CAR status to “BUSY” and then back to “IDLE”.

The CC accepts but ignores this command.

The behavior of the DC is documented in [18].

2-3.4.1.6 GUIDE

On receipt of this command, the NIRI IS will insure that an observation is not taking place. It will then set a flag as a reminder that guiding is taking place (as this information may affect its interaction with any on-instrument wavefront sensor). The sequencer will also forward this command on to the CC and DC. The “guideC” CAR record will be changed to “BUSY” and will change to “IDLE” when all the mechanisms are tracking within specification.

The CC accepts but ignores this command.

The behavior of the DC is documented in [18].

2-3.4.1.7 ENDGUIDE

On receipt of this command, the NIRI IS will insure that an observation is not taking place. It will then reset the guiding flag to indicate guiding has stopped. The sequencer will then forward this command on to the CC and DC. The “endGuideC” CAR record will be changed to “BUSY” and then back to “IDLE”.

The CC accepts but ignores this command.

The behavior of the DC is documented in [18].

2-3.4.1.8 OBSERVE

On receipt of this command, the NIRI IS will insure that an observation is not taking place and then forward the command to the DC. The IS will reject the command if its control software or the CC control software is still configuring its mechanisms. If accepted, the “observeC” CAR record will be changed to “BUSY”. It will go back to “IDLE” when the data set observation completes (or “ERR” if the observation was not successful).

The CC will not respond to this command.

The behavior of the DC is documented in [18].

2-3.4.1.9 ENDOBSERVE

On receipt of this command, the NIRI IS will forward the command to the DC. No changes will be made to the “observeC” CAR record.

The CC will not respond to this command.

The behavior of the DC is documented in [18].

2-3.4.1.10 PAUSE

On receipt of this command, the NIRI IS will forward the command to the DC.

The CC will not respond to this command.

In the DC, this will cause the observeC CAR record to change to “PAUSED”. Further details of the behavior of the DC is documented in [18].

2-3.4.1.11 CONTINUE

On receipt of this command, the NIRI IS will forward the command to the DC. The command will be rejected if an observation has not been paused. If successful, the “observeC” CAR record will be changed to “BUSY”.

The CC will not respond to this command.

The behavior of the DC is documented in [18].

2-3.4.1.12 STOP

On receipt of this command, the NIRI IS will forward the command to the DC. The command will be rejected if an observation is not being made. If successful, the “observeC” CAR record will be changed to “IDLE”

The CC will not respond to this command.

The behavior of the DC is documented in [18].

2-3.4.1.13 ABORT

On receipt of this command, the NIRI IS will forward the command to the DC. If successful, the “observeC” CAR record will be changed to “IDLE”.

The CC will not respond to this command.

The behavior of the DC is documented in [18].

2-3.4.1.14 PARK

On receipt of this command, the NIRI IS will insure that an observation is not taking place and then forward the command to the DC and the CC.

The CC will move its mechanisms to a configuration in which it can safely be switched off. It will utilize all the low-level CAD records dedicated to parking the individual mechanisms. The behavior of the DC is documented in [18]. The “parkC” CAR record will go “BUSY” while the mechanisms are being parked and go to “IDLE” if completed successfully or to “ERR” if any mechanism fails to park successfully.

The behavior of the DC is documented in [18].

2-3.4.1.15 DIAGNOSE

On receipt of this command, the NIRI IS will insure that an observation is not taking place and then forward the command to the CC.

The CC will perform self tests to ensure that its subsystems are healthy. The “applyC” CAR record will go “BUSY” while the tests are being carried out and go to “IDLE” if completed successfully or to “ERR” if any mechanism failed to diagnose successfully.

The DC will not respond to this command.

Note: unlike the TEST command, the DIAGNOSE command will move mechanisms, as necessary to diagnose problems.

2-3.4.2 General Engineering Commands2-3.4.2.1 REBOOT

On receipt of this command, the NIRI control software will reboot its IOC and go through the same start-up procedure it would normally do when first switched on. It will reboot into simulation mode NONE and debug mode NONE. The prototype and simulator will reboot into simulation mode FULL and debug mode NONE. There is no CAR record associated with this command because connection with the IOC will be lost while it reboots.

This command will also reboot the OIWFS components controller.

2-3.4.2.2 DEBUG

On receipt of this command, the NIRI control software will change to the given debug mode (NONE, MIN or FULL). This mode determines the amount and frequency of information logged by the NIRI control software, as described in the “Debugging Modes” section of ICD 1a, [24]. The “debugC” CAR record will go briefly “BUSY” and then back to “IDLE”.

2-3.4.2.3 FLUSH

This command will be forwarded to the DC.

The CC does not recognize this command.

On receipt of this command, the DC control software will transmit whatever is contained within its data buffers to the DHS data storage server for storage to disk. The action can be used for recovering data after a software problem. The “flushC” CAR record will go “BUSY” while the data are being transmitted and switch to “IDLE” if the data transfer succeeded or “ERR” if it failed. The behavior of the DC is documented in [18].

2-3.4.2.4 (Temperature controller commands)

TBD

2-3.4.3 Wheel and Wheel-like Mechanisms

NIRI will have two filter wheels, a pupil-mask/filter wheel, a focal plane mask and, two beam steerers. The following commands will act on all of the above mentioned mechanisms:

Each component can be set individually with the following commands:

2-3.4.3.1 Select position

On receipt of this command, the NIRI CC control software will:

Ensure an observation is not taking place.
 Translate desired position name into an engineering position with the aid of a look-up table.
 Look up the expected value for the Hall-effect sensors at the desired position location.
 Move the mechanism to the engineering position corresponding to the desired position.
 Verify that the expected sensor value matches the measured value.

2-3.4.3.2 Park mechanism

On receipt of this command, the NIRI CC control software will:

Ensure an observation is not taking place.
 Move the mechanism to its parking position.
 Verify that the expected sensor value matches the measured value.

2-3.4.3.3 Datum mechanism.

On receipt of this command, the NIRI CC control software will:

Ensure an observation is not taking place.
 Move the mechanism and search for its datum.
 Set the hardware datum position.

2-3.4.3.4 Set filter wheel position in engineering units.

On receipt of this command, the NIRI CC control software will move the mechanism to the specified engineering position. The Hall-effect sensors will not be used for confirmation.

2-3.4.3.5 Redatum mechanism.

On receipt of this command, the NIRI CC control software will:

Ensure an observation is not taking place.
Move the mechanism and search for its datum.

The hardware datum position will be left unchanged. This is a quick test that will detect a loss of positioning. It will detect arbitrarily large positioning errors, but will fail to detect small positioning errors.

2-3.4.3.6 Diagnose mechanism

On receipt of this command, NIRI will:

Ensure that an observation is not taking place.
Move the mechanism and attempt to verify that the motors and Hall-effect sensors are operating.

This is a quick test that will detect a loss of positioning. It will detect arbitrarily large positioning errors, but will fail to detect small positioning errors. If an error is detected, the diagnose command will fail and will return an error code in its CAR record.

2-3.4.4 Focus and offset commands

NIRI will have a focus stage, a window cover, and a pupil viewer. The following commands will act on the above mentioned mechanism:

2-3.4.4.1 Select position

On receipt of this command, the NIRI CC control software will:

Ensure an observation is not taking place.
Move the focus mechanism to the engineering position corresponding to the desired focus.
Compute the expected values for the Hall-effect sensors.
Verify that the final hall effect sensors match the expected values.

2-3.4.4.2 Park mechanism

On receipt of this command, the NIRI CC and control software will:

Ensure an observation is not taking place.
Move the focus mechanism to its parking position.

2-3.4.4.3 Datum focus

On receipt of this command, the NIRI CC and control software will:

Ensure an observation is not taking place.
Move the mechanism and search for its coarse datum.
Move the mechanism and search for its secondary datum (if necessary).
Move the mechanism and search for its datum.
Record the new datum position and use this to update the position to engineering position look-up table for the mechanism.

2-3.4.4.4 Redatum focus.

On receipt of this command, the NIRI CC and control software will:

Ensure an observation is not taking place.
Move the mechanism and search for its coarse datum.
Move the mechanism and search for its secondary datum (if necessary).
Move the mechanism and search for its datum.

The hardware datum position will be left unchanged. This is intended for detecting small positioning errors, before the errors have a chance to accumulate. It will underestimate the size of the error if the positioning error is large.

This command is not supported by the pupil viewer, because the pupil viewer uses a different home finding algorithm.

2-3.4.4.5 Diagnose filter wheel

On receipt of this command, the NIRI CC and control software will:

Ensure that an observation is not taking place.
Move the mechanisms and attempt to verify that the motors and Hall-effect sensors are operating.

This is a quick test that will detect a loss of positioning. It will detect arbitrarily large positioning errors, but will fail to detect small positioning errors. If an error is detected, the diagnose command will fail and will return an error code in its CAR record.

2-3.5 Detailed Status information

Details of these records are still TBD.

2-3.5.1 General records

2-3.5.1.1 name

This record contains a string constant describing the name of the instrument. For NIRI this record will contain "NIRI".

2-3.5.1.2 state

This record will take the values BOOTING, INITIALISING, RUNNING and CONFIGURING³ and reflect the state of the NIRI control software while it is initializing. The activities corresponding to these states are

- BOOTING — the state during IOC Init and while any database default values are being set.
- INITIALISING — the NIRI control software will be checking its own hardware and reading its hardware configuration files during this phase.
- RUNNING — the NIRI control software is ready to accept commands.
- CONFIGURING — the NIRI control software is ready to accept commands but it is busy configuring and unable to make observations.

3. Currently unimplemented

2-3.5.1.3 health

This record will reflect the overall health of the NIRI control software (including the components controller and the detector controller). Expected values are GOOD, WARNING or BAD. If the overall health is BAD then the NIRI control software will be unusable. If the Health is set to WARNING then the NIRI control software will be able to execute some functions but not perhaps to specification.

2-3.5.1.4 historyLog⁴

The record used to contain messages to be recorded in the history log.

2-3.5.2 Wheel records

These are TBD.

2-3.5.3 Focus/Offset records

These are TBD.

2-3.5.4 Environment Records

These are TBD.

2-3.6 Alarm conditions

The following situations will generate an alarm or warning:

- Hall effect sensor disabled
- Motor-control module fault
- Diagnose failed
- Redatum failed
- Datum failed
- Move-confirmation failed
- Temperature changing
- Large temperature gradient
- Limit sensor reached
- Motion out of valid range

2-3.7 Debugging and Maintenance

This interface can be tested and debugged by means of an engineering user interface supplied as part of the NIRI.

2-3.7.1 Debugging Modes

The NIRI control software will have the standard debugging modes described in ICD 1a, [24].

2-3.7.2 Calibration

If a Hall effect sensor fails or if a mechanism must be disassembled for maintenance, the sensors will need to be recalibrated. A separate program will be provided for this purpose. Calibration procedures will be described in a separate document [5].

4. Currently unimplemented

2-3.8 Simulation

The Gemini simulation modes are described in ICD 1a, [24]. The NIRI control software will use the FAST and FULL modes, and by default it begins in “NONE” mode. VSM mode may be supported, if time permits.

2-4.0 Safety Issues

It will be necessary to avoid moving NIRI components while the mechanical components are not at thermal equilibrium. Records will be provided to provide a temperature interlock. Details are described in [6].